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1 CHATURANGA v1.0.5.2 – REMAINING FEATURES BUILD LIST
2 For Claude AI. Vanilla JS/HTML. No frameworks. Match existing dark gold aesthetic.
3 Existing globals: ChaturangaELO, ChaturangaClock, game (Game instance), ui.js functions.
4 localStorage prefix: chaturanga_*
5
6
7 1. ADAPTIVE BOT – js/adaptive-bot.js (~20 lines)
8
9 - Read last 10 bot games from chaturanga_elo_v2 history
10 - Win rate >70% → next bot ELO += 100. Win rate <30% → next bot ELO -= 100
11 - Clamp result to 100-1000
12 - Store adjusted ELO in localStorage key: chaturanga_adaptive_elo
13 - On game start, if adaptive mode enabled, call setBotConfig(adaptiveELO) before board init
14 - Show small toast: "Difficulty adjusted ↑" or "↓" using existing ui toast pattern
15 - Expose: AdaptiveBot.getNextELO(), AdaptiveBot.setEnabled(bool)
16
17
18 2. CRITICAL MOMENT DETECTION – seer additions
19
20 Files: new js/analysis-worker.js + additions to seer.html/seer.js
21
22 analysis-worker.js (Web Worker):
23 - Receives: { positions: [{fen-like state, moveHistory, playerIndex}], botELO: 400 }
24 - For each position, runs ChaturangaTieredBot.getMove(game, 400)
25 - Compares material delta of bot's suggested move vs actual move made
26 - If delta difference >= 3 points (Ashwa=3, Ratha=5, Danti=3, Raja=∞), flag as missed win
27 - Returns array of { moveIndex, actualMove, betterMove, deltaGap, pieceType }
28
29 seer.js additions:
30 - After game loads in Seer, spawn analysis-worker with blunder positions
31 - In move log, add gold lightning ⚡ icon next to blunder moves
32 - On click: expand panel showing "Bot found [piece] to [square] – would have won [X] material"
33 - Draw gold arrow on board at the missed move square
34 - Section heading: "Turning Points" above the move log
35
36
37 3. ITIHASA – itihasa.html (replay theater)
38
39 Standalone page. Reads Granth vault from localStorage (chaturanga_granth_vault).
40
41 Layout (dark parchment, Cinzel headings matching elo-profile.html style):
42 - Left panel: vault list – game date, opponent, result, inscription count
43 - Right panel: replay board + scrubber timeline at bottom
44 - Board: read-only 8x8 render (copy board CSS from game.html, strip interactivity)
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46  Replay engine:
47  - Parse moveHistory array from saved game object
48  - Scrubber: range input 0-moveHistory.length, prev/next buttons, auto-play toggle (1 move/sec)
49  - On scrub to move N: rebuild board state by replaying moves 0..N from starting position
50  - When scrubber passes a move that earned a Granth inscription: fire inscription card overlay (reuse Gran
51  th SVG card renderer, call ChaturangaGranth.renderFromGame partial)
52  - Show move notation in sidebar as scrubber advances
53
54  Nav link: add "Itihasa" to game.html nav and granth.html nav.
55
56  4. MOVE NOTATION + SHARE URL – js/cn-encoder.js + additions
57
58  cn-encoder.js:
59  - Define Chaturanga Notation (CN) format: R:e1-e5 B:e8-e5x A:d4-f5 Y:pass
60    Player initials R/B/G/Y, from-to, x=capture, p=promotion, f=forfeit
61  - encodeCN(moveHistory) → compact string
62  - decodeCN(string) → moveHistory array
63  - toShareURL(moveHistory) → base64 encode CN, append as ?game=BASE64 to current URL
64  - fromShareURL() → check window.location.search for ?game=, decode if present
65
66  Seer integration:
67  - Add "Share game" button in game-over overlay (copy existing button style)
68  - On click: call toShareURL, copy to clipboard, show "Link copied" toast
69  - On seer.html load: call fromShareURL(), if found auto-load the game into replay
70
71
72  5. DICE PROBABILITY HEATMAP – seer.html addition
73
74  New section in Seer post-game panel: "Pāśaka Chronicle"
75
76  - Count face frequency (1-6) per player from moveHistory dice rolls
77  - Expected: each face = totalRolls/6 (16.7%)
78  - Render SVG bar chart: grouped bars per face value, 4 player colours
79  - Overlay expected distribution as a thin gold horizontal line per bar group
80  - If any player's distribution is >2 standard deviations from expected: show annotation "Auspicious Misfo
81  rtune – [Player] rolled [face] [N] times, expected [X]"
82  - All SVG, no external chart libs
83
84  6. OPENING LIBRARY – js/openings-data.js + ui additions
85
86  openings-data.js:
87  - Object mapping first-4-halfmove sequences to opening names + descriptions
88  - ~20 openings. Examples:
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89   Bhishma's Wall: Rathas advance to d1,e1 files in moves 1-2
90   Arjuna's Surge: Both Ashwas advance on moves 1-2
91   Karna's Gambit: Nara sacrificed on move 1 to open file
92   Drona's Formation: Danti to centre diagonal move 1
93   (build out 16 more in same pattern)
94   - Each entry: { id, name, description, moves: ['R:a1-a4','B:...'], granth_achievement_id }
95
96   ui.js addition:
97   - After move 4, check moveHistory[0..3] against openings-data patterns
98   - If match: flash small banner top-center of board: "[Opening Name]" (2s, then fade)
99   - Store matched opening in current game state
100  - Pass to Granth renderer so it appears in post-game manuscript
101  - Add 20 opening entries to chatbot-data.js (Acharya can explain each)
102
103  =====
104  7. BOARD THEMES AS ACHIEVEMENTS – additions to game.html + granth.html
105  =====
106  No new files needed. Extend existing board-theme selector.
107
108  5 locked themes + their unlock conditions:
109  - Kurukshetra at Sunset: complete Campaign Day 13 (check kurukshetra progress in localStorage)
110  - Indraprastha Marble: reach Arjuna rank in Akhara (check akhara_rank localStorage key)
111  - Ashoka's Pillar: 30-day Shubh streak (check shubh_streak key)
112  - Shakuni's Court: use Pāśa mechanic in 10 games (check pasha_count key)
113  - Midnight Ashtāpada: play 5 games between 23:00-03:00 IST (check timestamps in elo history)
114
115  In game.html board-theme selector:
116  - Locked themes show padlock icon + one-line unlock hint
117  - On Granth achievement grant: call unlockBoardTheme(themeId) which sets chaturanga_unlocked_themes in localStorage
118  - On unlock: brief celebration animation (gold shimmer over board for 1.5s using CSS keyframe)
119  - Theme CSS: add 5 new :root overrides in themes.css with --board-light, --board-dark, --board-border variables
120
121  =====
122  8. LIVE SPECTATOR MODE – server-ws.js + lobby.html + game.html
123  =====
124  server-ws.js:
125  - Add spectators: Set to each room object (alongside players: Map)
126  - New WS message type: join-spectate { roomId }
127  - Server adds socket to spectators set, sends current game-state snapshot
128  - All game-state broadcasts go to players AND spectators
129  - Server never forwards move/roll messages from spectator sockets (guard by checking spectators set)
130  - Broadcast spectator-count { count } to all in room when spectators set changes
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132 lobby.html:
133 - Add "Watch" button on each room card alongside "Join"
134 - Show spectator count badge on room card if count > 0
135 - Watch button sends join-spectate message
136
137 game.html / ws-client.js:
138 - If connected as spectator: disable Roll button, disable board click/drag
139 - Show "Watching" banner in place of player controls
140 - Spectator chat sidebar: simple textarea + send, WS message type spectator-chat
141
142
143 9. REMATCH + REVENGE – server-ws.js + ui.js
144
145 Already: rematch-vote WS message type exists in ws-client.js. Just needs wiring.
146
147 server-ws.js handler for rematch-vote:
148 - Track votes per room in rematch_votes Map
149 - When all players vote: broadcast rematch-start { sideSwap: true }
150 - Side swap: rotate player colour assignments by 1 position
151 - Reset board, clear old game state, start fresh
152
153 ui.js game-over overlay additions:
154 - "Rematch" button → sends rematch-vote { type: 'rematch' }
155 - "Revenge" button → sends rematch-vote { type: 'revenge', preferredColor: currentColor }
156 - Show head-to-head record: read localStorage key chaturanga_h2h_{opponentId}
157   Format: "You: 3 wins · Opponent: 5 wins"
158 - Update h2h record on game-over: ChaturangaH2H.record(opponentId, won)
159 - Waiting state: "Waiting for opponent..." spinner after voting
160
161
162 10. SAMARA SHASTRA – samara.html (pre-battle briefing)
163
164 Standalone overlay page OR modal that fires before any game loads.
165
166 Reads three data sources:
167   A) Panchanga: import today's tithi/nakshatra/vara/yoga from panchanga.html logic
168     (copy getPanchangaToday() function, it already exists in panchanga.html)
169   B) Granth vault: last 3 inscriptions earned (tactical pattern player uses most)
170   C) Vyuha selection: player's chosen formation from current game setup
171
172 Generates briefing text by template:
173   "Today is [vara] under [nakshatra] – [yoga adjective].
174   Your Granth records [N] [top inscription name] inscriptions.
175   The [formation name] opens your [formation strength].
176   [Tactical advice combining all three]."
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177
178 Mapping table needed (hardcode in samara.html):
179 nakshatra → tactical advice string (27 entries)
180 vara → piece modifier hint (7 entries)
181 inscription → tactical pattern name (10 entries, from granth.html)
182 formation → strength description (6 entries, from vyuha-data.js)
183
184 Display:
185 Full-screen dark overlay, Cinzel font, gold text
186 Drona avatar (SVG, simple silhouette – reuse existing if present)
187 Briefing text types in character by character (typewriter effect, 40ms/char)
188 "Enter battle" button dismisses and loads game
189 Skip link bottom-right for repeat players
190
191 Trigger: add "Seek counsel" button to game setup screen in game.html
192 OR auto-show on first game of the day (check localStorage date key)
193
194

195 11. SANGRAM LIVE DHARMA BAR – additions to game.html + dharma-engine.js
196

197 Currently: dharma computed only post-game in dharma-engine.js.
198 Goal: live animated bar updating after every move.
199
200 dharma-engine.js additions:
201 - Export getDharmaSnapshot(moveHistory) → { scores: [R,B,G,Y], events: lastEvent }
202 - Compute incrementally: on each move append to running dharma state
203 - Events that change dharma (already defined in engine, just expose live):
204 sacrifice (+), forfeit (-), consecutive captures (+), raja capture (climax)
205
206 game.html additions:
207 - Add dharma bar element below/beside the existing score/material bar
208 - 4 coloured segments (R/B/G/Y) proportional to dharma scores, animated width transition 0.4s
209 - After each move: call getDharmaSnapshot, update bar widths
210 - On significant dharma event (delta > 15): flash brief annotation above bar
211 e.g. "Ratha sacrificed – Dharma rises"
212 - On Raja capture: full-width gold pulse animation on the bar (CSS keyframe 1s)
213
214 Visual spec:
215 Bar height: 6px, same width as board
216 Segment colours match player colours (red/blue/green/yellow at 0.7 opacity)
217 Background: rgba(255,255,255,0.04)
218 Annotation: small italic text, 1.5s then fade, positioned above bar centre
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220

221 12. WEEKLY YUDDHA LEADERBOARD – server.js + lobby.html

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223 server.js additions:
224 - Separate in-memory ledger: weeklyELO Map { playerId → deltaELO this week }
225 - On ranked game-over: POST /api/weekly-yuddha { playerId, delta }
226   Server adds delta to weekly ledger
227 - GET /api/weekly-yuddha returns top 10 { playerId, name, weeklyDelta, rank }
228 - Scheduled reset: check every hour if Sunday midnight IST has passed
229   On reset: grant Granth achievements to top 3, clear weekly ledger
230   Achievements: Saptaha Vijeta (1st), Agrani (2nd), Yoddha Shrestha (3rd)
231 - Persist weekly ledger to weekly-yuddha.json (same pattern as leaderboard.json)
232
233 lobby.html additions:
234 - Leaderboard panel in lobby sidebar
235 - Fetch /api/weekly-yuddha on load, render top 10 rows
236 - Show countdown to next Sunday reset (compute from current IST time)
237 - Top 3 rows get gold/silver/bronze left-border accent
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240 13. OFFLINE PWA IMPROVEMENTS – sw.js + nav additions
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242 sw.js:
243 - Add offline.html to cache on install
244 - Add Background Sync registration for Shubh solve events
245   Queue: shubh-sync-queue in IndexedDB, retry on reconnect
246 - Fetch handler: on network fail, return offline.html for navigation requests
247
248 offline.html (new file):
249 - Simple dark page matching game aesthetic
250 - Message: "You are offline – AI practice and daily puzzles are still available"
251 - Two buttons: "Play vs AI" (links to game.html) and "Daily Shubh" (links to shubh.html)
252 - No external font requests (use system serif fallback)
253
254 Nav connectivity dot (add to all pages' nav bar):
255 - Small circle element id="connectivity-dot" in top nav
256 - green = online, amber = reconnecting (show on WS disconnect), grey = offline
257 - JS: window addEventListener online/offline + WS onclose/onopen events
258 - CSS: 8px circle, position absolute top-right of nav, transition color 0.3s
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261 14. CLAN SYSTEM (SENA) – sena.html + lobby.html + server.js
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263 sena.html (new page):
264 - Create Sena: name input + choose Vyuha emblem (6 Vyuha icons from vyuha-data.js)
265 - Generate 6-char invite code (random, stored as clan ID)
266 - Join Sena: enter invite code
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267 - Sena panel: list of members with their ELO and recent Granth inscriptions
268 - Leave / Disband buttons
269
270 localStorage (clan data, no server needed for MVP):
271 - chaturanga_sena: { id, name, vyuha, members: [{id, name, elo}], inviteCode }
272 - Member ELO pulled from their local chaturanga_elo_v2 (shared via invite flow)
273
274 lobby.html:
275 - Show Sena badge next to player name if in a clan
276 - Sena leaderboard tab: weekly wins aggregated across members
277
278 server.js (optional server-side for shared clans):
279 - POST /api/sena/create, /api/sena/join, GET /api/sena/:id
280 - If server unavailable: graceful localStorage-only fallback
281

282
283 BUILD ORDER RECOMMENDATION

284
285 Session 1 (single-player depth, already started):
286 → 1. Adaptive Bot (20 lines, immediate)
287 → 2. Critical Moments (closes play-to-learn loop)
288 → 3. Itihasa (replay theater, uses existing Granth data)
289

290 Session 2 (distribution + discovery):
291 → 4. Share URL / CN (viral)
292 → 6. Opening Library (vocabulary layer)
293 → 5. Dice Heatmap (validates frustration, quick SVG build)
294

295 Session 3 (visual rewards + feel):
296 → 7. Board Themes (zero new infra, immediate delight)
297 → 11. Sangram Dharma Bar (live emotional register, small addition)
298 → 10. Samara Shastra (biggest atmosphere payoff)
299

300 Session 4 (multiplayer layer):
301 → 8. Spectator Mode (social proof)
302 → 9. Rematch + Revenge (strongest post-loss retention)
303 → 12. Weekly Yuddha (week-long tension)
304

305 Session 5 (platform completion):
306 → 13. Offline PWA (India market)
307 → 14. Clan System (largest effort, highest long-term stickiness)
308